

RAMAN PANDEY

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EDUCATION

University of New Mexico Ph.D. Electrical & Computer Engineering (advisor: M. Osiński) Aug. 2026 – Present
University of New Mexico M.S. Electrical & Computer Engineering (QIS concentration) Aug. 2025 – Dec. 2026 (exp.)
Relevant courses: Quantum Error Correction, Quantum Optics, Introduction to Quantum Information Science (QIS), Platforms of Quantum Technology, Deep Learning
University of New Mexico B.S. Computer Engineering; B.S. Astrophysics Aug. 2021 – May 2025
Relevant courses: Instrumentation, Microprocessors, Intermediate Logic Design, Computational Physics, Modern Physics, Radio Astronomy

RESEARCH EXPERIENCE

Center for High Technology Materials, Graduate Research Assistant (undergraduate 2023–25) **2023 – Present**

- Area lead for **Clock and System-Level Engineering** on the NSF NQVL QCAP photonic quantum computing project (feedforward circuits, physical clock rate, QEC resource overhead, systems integration); sub-area lead for quantum-dot electrical driving and detector data processing.
- Developed (starting as a senior design project) readout systems for **superconducting nanowire single-photon detectors (SNSPDs)**, converting analog pulses through a 50 dB low-noise amplification chain and comparator digitization into time-tagged photon events with 20 ps timing resolution.
- Built time-resolved photon-detection pipelines on a **Trenz TE0712 (Artix-7) FPGA module**, implementing 20 ps-bin time-to-digital conversion and asynchronous counting at 200 MHz with ~ 6 ns dead time (~ 166 Mcps theoretical maximum count rate), validated against a commercial time tagger (Swabian Instruments).
- Built an **LTspice electro-equivalent model of NbTiN SNSPDs** to study latching suppression: showed that a feedback-bias scheme, unlike passive shunts, widens the latch-free load range (63 to 97 Ω at 27 nH) with $\sim 25\%$ faster recovery, and derived its controller-latency limit (≤ 50 ps at 27 nH).
- Designing an FPGA-clocked **electrical excitation scheme for III-V p-i-n quantum-dot single-photon emitters** (DC bias plus 100 ps injection pulses at 80 MHz as the simulated design point), targeting $g^{(2)}(0) < 0.5$ purity in non-cryogenic, in-plane SiN-integrated operation.
- Designing dynamic gated-overvoltage biasing for **single-photon avalanche diodes (SPADs)**, synchronizing detector gates to photon-arrival windows to suppress afterpulsing and dark counts.
- Engineered system-level architecture for a **Gaussian Boson Sampling** platform, integrating quantum-dot emitters, an 8-mode, 28-MZI Clements **Mach-Zehnder interferometer** mesh (all 64 parameters of the 8×8 unitary), SPADs, and FPGA control of the electrical phase shifters under a unified timing framework.
- Prepared the equipment list, block diagrams, and six-year NSF budget justification (~ 23 vendor quotes) for four boson-sampling testbed configurations in the QCAP implementation-phase proposal.
- Mentored undergraduate and high-school students on FPGA/VHDL design, single-electron devices, and device simulation.

TEACHING EXPERIENCE

UNM Electrical & Computer Engineering, Teaching Assistant **Jan. 2025 – May 2026**

- Instructed 80+ students across 5 lab sections over 3 semesters in **FPGA and embedded-systems labs** (ECE 238L, ECE 344L) using Vivado/VHDL (Artix-7) and ARM Cortex-M (TM4C123), covering FSMs, GPIO, interrupts, timers, PWM, ADC, UART/SPI/I2C, and oscilloscope and JTAG debugging.
- Implemented an automated hex-code assignment system that eliminated duplicate lab submissions.

PROJECTS

qd-photon-sim: Quantum-Dot Single-Photon Source Simulator (2026 – Present, github.com/Alpharceus/qd-photon-sim): Python device simulator (190+ automated checks) for electrically driven quantum-dot sources, covering biexciton-cascade spectral filtering, DBR microcavity design, thermal carrier escape, and pulsed-injection photon statistics; maps $g^{(2)}(0)$ and collected photon rate across 230–300 K operating points and drive mechanisms.

MZIMesh: Photonic Mesh and Boson Sampling Simulator (2026 – Present): Julia simulator for Clements and Reck interferometer meshes up to 10 modes; Clements decomposition of arbitrary unitaries (round-trip error $< 10^{-12}$), Fock-input boson sampling via Ryser permanents, and a Makie dashboard; 680+ tests.

SKILLS

Programming: Python, Julia, C/C++, VHDL, Verilog, MATLAB, ARM Assembly

Quantum & Photonic Systems: Single-photon sources, single-photon detectors (SNSPDs, SPADs), quantum dots, Gaussian Boson Sampling (GBS), MZI meshes, photonic quantum computing, quantum error correction

Hardware & Embedded: FPGA design (Vivado, ModelSim), ARM Cortex-M microcontrollers, time-to-digital conversion

Tools: LTspice, PSPICE, LabVIEW, Multisim, Git, oscilloscopes, JTAG debugging